

Identities, Transformations & Equation Structure

IOQM Grade 9 - Assimilation + First-Step Pack

For the learner who can manipulate algebra but does not yet consistently choose the representation that makes the requested target smallest.

Validated anchors: IOQM-2025-Q01; IOQM-2025-Q21; IOQM-2024-Q05; IOQM-2024-Q11. Exact historical wording remains under source-paper custody.

Concept map

Governing idea: read the target first. Choose factor, expand, substitute, symmetric reconstruction, power reduction, or an equivalence check because that representation makes the requested information smaller.

Reconnect

$49^2 - 31^2$: factor rather than square both numbers.

$u+v=10$, $uv=21$, asked for u^2+v^2 : reconstruct a symmetric identity.

$x^2+x=1$ with a high-power target: use $x^2=1-x$ as a rewriting rule before solving x .

Identity vs relation

An identity is true for every permissible input. An equation such as $x^2+x=1$ is true only on its solutions, but there it becomes a valid rewriting rule. This distinction prevents careless transformations.

Substitution

A substitution is useful when it compresses a repeated block, e.g. $t=x+1/x$. Do not introduce a new symbol if it does not reduce structure or if it hides a domain condition.

Equivalence matters

Squaring can add candidates.

Dividing by an expression can lose the zero branch.

Clearing denominators requires excluding zero denominators first.

A legal algebraic step is not automatically an equivalent step.

Try before hints

Given $a+b=12$, $ab=20$, find a^2+b^2 . H1: symmetric target. H2: use $(a+b)^2$. H3: 104.

Given $x^2=3x-1$, reduce x^5 to $Ax+B$. H1: use the relation repeatedly. H3: $x^5=55x-21$.

First-Step Reference

difference of squares $\rightarrow$ factor?

repeated composite block $\rightarrow$ substitute?

symmetric target $\rightarrow$ sum/product identity?

high power + low-degree relation $\rightarrow$ reduce powers?

square/divide/clear denominator $\rightarrow$ what domain/equivalence check?

H0 mastery

1) Evaluate $1007^2 - 993^2$.

2) $a+b=13$, $ab=36$: find a^3+b^3 .

3) $x^2=2x+5$: express x^5 as $Ax+B$.

4) $p-q=4$, $pq=21$: find p^2+q^2 .

5) Solve $\sqrt{3x+4}=x+2$ with all checks.

6) If $u^2+u+1=0$, determine u^{2026} .

Adopt: read the target first; then choose the representation.